

Macroeconomic Theory II (1412)

A simple dynamic model (backward looking) *IS-LM* model

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Scan the QR code to visit the MT2 GitHub repository



Exercise 1 Assume the following closed economy, total demand is given by:

$$D_t = C_t + I_t + G_t$$

The ad-hoc behavioural equations for private consumption and investment are given by:

$$C_t = \bar{C} + c(Y_{t-1} - T_{t-1})$$

and

$$I_t = \bar{I} + \alpha Y_{t-1} - b i_{t-1}$$

In your solution assume the following parameterization $\bar{C} = 0.6, \bar{I} = 0.2, c = 0.5, \alpha = 0.1, b = 0.1$. Also assume that the fiscal authority follows a balanced budget

$$G_t = T_t = 1.7$$

while the monetary authority sets the nominal interest rate

$$i_t = 0.04$$

Solve for the steady state. Create a matlab function that find the steady state solution of this model (i.e., drop t).

Exercise 2 Assume that the closed economy starts off the steady state equilibrium, i.e., the initial value of output is:

$$Y_{-1} = 0.9Y^s$$

where Y^s denotes the steady state output which is computed in exercise 1.

1. Create a matlab function that computes the dynamic equilibrium transition from Y_{-1} towards the steady state output, Y^s . (**Create a vector with the time path of the endogenous variables of the model.**)
2. Plot in a single plot three subplots with the dynamic path of the endogenous variables of the model.

3. Repeat the above for an economy with $c = 0.4$ and $c = 0.7$.
4. Present in the same plot the dynamic transition of each economy, i.e., for $c = [0.4, 0.5, 0.6]$

Exercise 3 Permanent Policy Changes

The economy starts from the steady state described in Exercise 1. Consider the following macroeconomic policy changes, each implemented in period $t = 1$ and analyzed separately. These changes are permanent, implying that the economy transitions to a new steady state.

1. **Increase in government spending.**
The fiscal authority increases government expenditures by 50%.
2. **Increase in taxes.**
The fiscal authority increases taxes by 30%.
3. **Monetary policy tightening.**
The monetary authority increases the nominal interest rate by 100 basis points (bps).

For each of the above policy changes:

- Compute the new steady state and illustrate it graphically, showing both the initial and the new equilibrium.
- Compute and present the transitional dynamics of the endogenous variables of the model.
- Suppose that employment is proportional to output:

$$L_t = \gamma Y_t$$

Compute and plot the transitional dynamics of employment, using $\gamma = 0.7$.

Exercise 4: Transitory Policy Changes

We now study transitory policy changes. The economy starts from the steady state computed in **Exercise**

1. Consider the following policy changes:

1. **Increase in government spending: Debt-financed**
The fiscal authority increases government expenditures by 10% in period $t = 1$, and thereafter G_t returns gradually to its steady-state value according to:

$$G_t = \bar{G} + \rho^G (G_{t-1} - \bar{G})$$

where $\bar{G}$ is the initial steady state value. Set $\rho^G = 0.5$.

Hint: In period $t = 1$, $G_1 = 1.1 \times G_0$.

Assume that the increase in government spending is financed through the issuance of public debt (debt financing).

2. **Increase in government spending: Balanced budget policy.**
Compare the above scenario with a balanced budget scenario, i.e., where the increase in government spending is fully financed by an equivalent increase in taxes.
In your reply (pdf), clearly present and explain the differences between the two scenarios and their macroeconomic implications (e.g., you may present the scenarios in the same plot).
3. **Increase in taxes**
The fiscal authority increases taxes by 10% in period $t = 1$. Taxes then return to their steady-state value following:

$$T_t = \bar{T} + \rho^T (T_{t-1} - \bar{T})$$

where $\bar{T}$ is the initial steady state value. Set $\rho^T = 0.5$.

4. Monetary policy shock.

The monetary authority increases the nominal interest rate by 75 basis points (bps) in period $t = 1$. The interest rate then gradually returns to its steady-state value according to:

$$i_t = \bar{i} + \rho^i (i_{t-1} - \bar{i})$$

where $\bar{i}$ is the initial steady state value. Set $\rho^i = 0.5$.

For each of the above policy experiments:

- Present the relevant impulse response functions (plots) and provide **very brief economic interpretations**. (Please avoid extensive use of AI tools; you may use them for proofreading.)
- As above, compute and plot the transitional dynamics of employment, using $\gamma = 0.7$.
- Finally, conduct a sensitivity analysis with respect to the persistence parameter of each policy shock. In particular, examine how different values of $0 < \rho^{\{G,T,i\}} < 1$ affect the dynamic responses of the economy over time.
- As a general guideline, first plot the path of the policy instrument (e.g., G_t , T_t , i_t), and then the implied equilibrium paths of the endogenous variables.